

Homework 2

due 01.11.2023, 14:00

Exercise 3.1 - Point to Point Communication (5 points)

Write a function that mimics MPI's broadcast function, but only uses point to point communication.

```
int my_Bcast_Loop(void *buffer, int count,
                  MPI_Datatype datatype,
                  int root, MPI_Comm comm)
{
    ...
}
```

Use a loop in which the root process sends the data to every other process. Write also a main function in which an array of 100000 doubles is set up and broadcasted both with the built-in and your own broadcasting routine. Demonstrate that the broadcast is correct and compare the timings between the two using 48 cores (two nodes on the `compute2011` partition on `stromboli`).

Exercise 3.2 - Tree (5 points)

Write another broadcast function `my_Bcast_Tree` and repeat the tasks from the previous problem. This time, exploit that many point-to-point communications can be done concurrently, and proceed in a tree like fashion. A possible algorithm works like this:

- Given `numprocs` processes, find the smallest integer k with $\text{numprocs} \leq 2^k$
- Loop $i = k, \dots, 1$
 - all processes with $\text{rank} \bmod 2^i = \text{root}$:
send data to processes with rank $\text{rank} + 2^{i-1}$ (if it exists).